The Evolution of Information Systems

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Information Science broadly refers to the intersection of technology and theory for the collection, movement, and storage of information for human use (Borko, 1968). This includes a spectrum of concepts that build from raw data to sophisticated wisdom (Buckland, 1991). The methods for managing information fall under the umbrella of information systems. Through time, information systems have developed and changed dramatically. Their evolution has had a significant impact on the timeline of human progress. The very first information systems resided solely in human minds. Prior to the invention of the written word, early advances in information systems stored valuable information with objects and symbolism. These methods eventually developed into vast collections of written knowledge housed in libraries. In the modern day, information systems have graduated to recognizable technologies like databases within computer systems. Their future is an open frontier filled with creative ideas and technological challenges. From early innovations like the printing press which radically increased the spread of information and literacy to modern-day digital information systems, each step forward has influenced human culture, knowledge, and interconnectedness on a global scale. Information systems are the backbone of how humans store, process, and share knowledge. Collectively, the evolution of information systems throughout time has contributed significantly to the advancement of human progress.

The emergence of early information systems predates the invention of written language by thousands of years. In the absence of written records, prehistoric humans relied on the only available storage medium of the time: human memory. The human capacity for collecting, storing, and transmitting vast amounts of information was crucial for survival and adaptation, enabling early humans to understand and navigate the world they lived in. Knowledge about food sources, seasonal patterns, predator behaviors, and the other people around them were

passed down from generation to generation. However, this transfer of knowledge was unsecured. Since much of life's essential information was not encoded in DNA, no human was born with an innate understanding of which plants were edible or poisonous, nor the diction and songs of their culture's language. Instead, each individual had to learn from the environment, from others, and from the oral traditions that shaped early societies. Oral traditions, storytelling, and verbal communication became the backbone of early human information systems (Kelly, 2015, p. 37). These practices provided a means of preserving and passing down critical knowledge. However, they were also inherently volatile. Information risked being permanently lost due to an individual's death. It could also be corrupted either deliberately or due to natural errors in human memory.

This reality pointed to a crucial gap in early information systems: the lack of a reliable and long-lasting storage mechanism for knowledge outside of the human mind. In response to this problem, early symbols and rudimentary forms of written communication began to emerge. The development of written symbols marked a pivotal moment in the history of information systems. The revolutionary use of imagination in symbolic thinking, being the ability to use one object, image, or entity to represent another, was foundational for the advancement of information storage (Vyshedskiy, 2022, p. 14-18.) One of the earliest and most enduring examples of successful, long-term information storage was found in the cave and rock art at sites such as Chauvet Cave in France, which date back over 30,000 years and have survived to the present day (Onians, 2018, p. 3-4). These artworks, while likely initially serving ritualistic or symbolic purposes, also functioned as a form of recorded knowledge, embedding information in a lasting, non-biological medium. Over time, these early information systems provided a foundation for the more sophisticated methods of information storage and transmission that

would follow. The movement away from relying solely on human memory to record information marked a fundamental shift that paved the way for the development of all modern information systems.

With the expanding institution of imaginative thinking came significant advances in the use of symbolism in early information systems. The need for growing societies to effectively manage their resources and track trade led to the creation of increasingly complex methods of record-keeping. As societies expanded beyond small, localized groups, it became essential to document and preserve transactions, allocate resources, and maintain an organized system of accounting that could span generations. The early innovations in symbolic representation provided the foundation for these systems long before the invention of alphabetic writing (Vyshedskiy, 2022). In ancient Mesopotamia, one of the earliest known examples of record-keeping involved the use of clay tokens. The tokens were crafted into specific shapes to represent different types of data, then stored in hollow clay containers for extended periods of time (Mattessich, 1989, p. 75-76). Mattessich (1989) describes how these containers served as early accounting tools, with the shapes of the tokens representing various entries that could be used to track financial records. For example, one container filled with tokens of different shapes might indicate an individual account's "assets or equity" (Mattessich, 1989, p.75). Across the world, other ancient civilizations were also developing their own methods of recording data and numbers. In the Andean regions of South America, the Incas and other civilizations devised an intricate system known as quipu. This system utilized knotted strings to record numerical information. According to Ascher (2005), quipu knots used color, position, and knot-type to represent "digits [and] powers of ten" (p. 100). Quipu, like the clay tokens of Mesopotamia, was an effective, symbolic system for tracking numerical information.

Succeeding early record systems, the invention of written language marked a monumental turning point in human history. As the complexity of societal structures grew, so did the need to create and store records of increasingly complex information. A new era of information systems had already begun when humans began to deviate from relying solely on oral traditions in favor of recorded data. While earlier methods of information storage allowed societies to manage essential data, they were limited in their ability to preserve complex, abstract information. Written language offered a solution to these limitations by providing a more reliable, long-term means of capturing, storing, and transmitting knowledge. An early precursor to alphabetic writing came in the form of pictographs, which were simple systems of symbols used to represent objects, animals, or concepts (Wan, n.d.). The ancient Egyptians, Sumerians, and Mayans were all among the first to develop and use pictographs. Phonetic systems, or written systems that broke down to represent sounds rather than specific ideas, later evolved from earlier written systems like these (Houston, 2003). One famous example of an early phonetic language is that of cuneiform, which remained in use for over 3,000 years in the ancient Middle East (Wojahn, 2004). The first written-information specialists were called scribes. The role of the scribe became increasingly important in ancient civilizations, as writing was a specialized skill that few could master. Scribes were employed to record economic transactions, preserve legal codes, and write religious texts onto clay tablets, a task that required years of training and expertise (Podany, 2012).

The creation of written information systems also led to a societal restructuring of access to knowledge. For the first time in human history, mass information repositories housed the world's collective achievements in one space, allowing scholars to study and learn from the works of their predecessors. Among the most remarkable of these were the ancient Library of

Alexandria and the Bodleian Library at Oxford, the latter of which still stands to this day (Krummel, 2002, p. 539). Libraries like these housed thousands of scrolls and documents containing the accumulated knowledge of various civilizations, with topics ranging from mathematics and astronomy to philosophy and medicine. Information repositories symbolized the transcendence of written language as an enduring means of preserving human thought and progress, ensuring that the intellectual endeavors of one generation could pass on to the next. However, the access to written knowledge was not equally distributed. In ancient civilizations, literacy was often confined to political and religious elites who controlled the resources necessary for maintaining libraries and archives. Writing became a tool of social stratification with information serving as a source of political and religious power (Kelly, 2015, p. 24-33). The "knowledge elite" not only controlled access to knowledge, but also held the power to interpret and alter it at their discretion (Kelly, 2015, p. 25). This power structure of exclusivity continued throughout the majority of human history. During this time, monks and scribes meticulously created few copies of existing texts by hand, a slow and labor-intensive process that maintained books and manuscripts as rare and valuable commodities (Robinet, et al., 2024).

But the slow, exclusive nature of information systems changed radically with the rise of modern information technologies. In the 15th century, the invention of the printing press by Johannes Gutenberg allowed for the mass production of books, making written knowledge more accessible than ever before (Krummel, 2002, p. 539). As Dittmar states, the invention of the printing press was "one of the most revolutionary inventions in human history" (p. 1133). Books could now be produced quickly and cheaply, making them available and affordable to those with lesser income and social influence. Over time, the production of printed books continued to improve and proliferate. By the industrial revolution, literacy only continued to rise (Krummel,

2002, p. 540-541). In 1938, information technology took another leap forward when Alan Turing's "Turing Machine" laid the groundwork for all modern computing. During World War II, the Turing Machine successfully cracked the Nazi Enigma code, transforming information systems forever and marking the beginning of the digital age (Kruh, 2006, p. 287-288). Less than 50 years later, the first mass-produced computers were being sold to consumers for personal use (Ensmenger, 2012, p. 764-765). Unlike previous approaches which relied on physical storage mediums, digital databases revolutionized how information could be accessed, changed, and copied in seconds. Not so unlike the symbols of the ancient past, binary computer technology allowed for series of ones and zeros to represent beautiful artworks and lengthy novels which could be stored on a flash drive of merely inches in size. With the rapid advancements in computer technology, the internet soon followed, creating a global network for the instantaneous exchange of information from anywhere in the world. Now, it was possible to access almost any information and learn about anything desired on the spur of a moment. Ensmenger reflects that "we are living through a technological revolution unprecedented in all of human history" (p. 757).

Looking to the future, information systems face both significant challenges and intriguing opportunities for technological advancement. While modern computers have become very powerful, they remain constrained by hardware limitations, processing speeds, and the ever-increasing energy demands of data storage and computation. As data-intensive fields like big data, artificial intelligence, and large-scale simulations continue to expand, these limitations raise concerns about the sustainability of traditional electronic systems (Buyya et al., 2024, p. 25). To address these challenges, quantum computing and chemical, DNA-based computing offer exciting potential solutions. Quantum computers, which utilize quantum bits or "qubits," have

the ability to process vast amounts of data simultaneously, enabling faster processing speeds and more energy-efficient computation (Buyya et al., 2024; Sood & Chauhan, 2024). However, quantum computing faces obstacles such as scalability and error correction, but it still holds the promise of transforming industries like cryptography, data analysis, and AI while reducing energy consumption (Tang et al., 2024, p. 133). In addition to quantum computing, DNA-based chemical computing offers a high-density, stable, and environmentally sustainable alternative to traditional electronic storage (Yang et al, 2024). Researchers are exploring how microfluidic technology can facilitate the reading, writing, and storage of data in DNA, providing a potential solution to the energy demands and storage limitations of current systems (Zhang, & Nguyen, 2017). Both quantum and DNA computing present technologically viable paths forward, offering the potential to overcome the current limitations of electronic information systems and revolutionize information technology yet again. Information systems of the future will have to seek more efficient, sustainable ways to manage the vast collection of humanity's data and knowledge.

Ultimately, human progress has advanced throughout time with the evolution of information systems. From oral traditions and symbolism to complex written languages and digital databases, each step of information systems' history has been essential in shaping how societies learn, communicate, and grow. Information systems have undeniably transformed nearly every aspect of modern life, from the way financial trade operates to how individuals connect with one another across the globe. They have enabled unprecedented access to information, facilitated learning, innovation, and efficiency. As societies continue to advance technologically, the role of information systems will only grow, driving further improvements in various sectors. Looking ahead, the potential for information systems to contribute to even

greater achievements remains promising. With the advent of newer technologies such as quantum and DNA-based chemical computing, the possibilities are limitless. As technology advances, information systems will undoubtedly become even more integral to addressing global challenges. The future holds opportunities for systems that support sustainability efforts, enable better resource management, and aid in knowledge acquisition. The continued development of information systems will shape how societies function, opening doors to new discoveries. The impact of these systems will likely increase, paving the way for a world that is not only more interconnected but also more informed, capable, and prepared to meet the demands of an ever-evolving global environment. Each day, new developments promise to redefine the possibilities for information systems, ensuring that knowledge can continue to be preserved, analyzed, and shared across generations in the modern world, regardless of income or social status, in ways that were once unimaginable.

References

Ascher, M. (2005). How can spin, ply, and knot direction contribute to understanding the quipu code? *Latin American Antiquity*, *16*(1), 99–111. https://doi.org/10.2307/30042488

Borko, H. (1968). Information science: What is it? *American Documentation, 19*, 3-5.

Buckland, M. K. (1991). Information as thing. *Journal of the American Society for Information Science*, *42*(5), 351–360. https://doi.org/10.1002/(SICI)1097-4571(199106)42:5<351::AID-ASI5>3.0.CO;2-3

Buyya, R., Ilager, S., & Arroba, P. (2024). Energy-efficiency and sustainability in new generation cloud computing: A vision and directions for integrated management of data centre resources and workloads. *Software, Practice & Experience*, *54*(1), 24–38. https://doi.org/10.1002/spe.3248

Ensmenger, N. (2012). The digital construction of technology: Rethinking the history of computers in society. *Technology and Culture*, *53*(4), 753–776. https://doi.org/10.1353/tech.2012.0126

Houston, S., Baines, J., & Cooper, J. (2003). Last writing: Script obsolescence in Egypt, Mesopotamia, and Mesoamerica. *Comparative Studies in Society and History*, *45*(3), 430–479. https://doi.org/10.1017/S0010417503000227

Kelly, L. (2015). *Knowledge and power in prehistoric societies: orality, memory, and the transmission of culture*. Cambridge University Press. *Information Processing & Management,* Volume 32, Issue 1, 1996, Pages 63-76, ISSN 0306-4573.

Kruh, L. (2006). Alan Turing and the invention of the computer. *Cryptologia*, *30*(3), 287-288.

Clementi, E. (2014). Evolution of computers and simulations: From science and technology to the foundations of society. *Science China. Chemistry*, *57*(10), 1317–1329. https://doi.org/10.1007/s11426-014-5155-2

Krummel, D. W. (2002). Libraries, history of. in J. R. Schement (Ed.), *Encyclopedia of Communication and Information* (Vol. 2, pp. 539-541). Macmillan Reference USA. https://link.gale.com/apps/doc/CX3402900154/GVRL?u=tamp44898&sid=bookmark-GVRL&xid=195892c3

Mattessich, R. (1989). Accounting and the input-output principle in the prehistoric and ancient world. *Abacus (Sydney)*, *25*(2), 74–84. https://doi.org/10.1111/j.1467-6281.1989.tb00222.x

Onians, J. (2018). Art, the visual imagination and neuroscience: The Chauvet Cave, Mona Lisa's smile and Michelangelo's terribilitá. *Cortex*, *105*, 182–188. https://doi.org/10.1016/j.cortex.2017.10.009

Podany, A. H. (2012). Scribes, ancient Near East. In *The Encyclopedia of Ancient History* (pp. 1–2). John Wiley & Sons, Inc. https://doi.org/10.1002/9781444338386.wbeah01171.pub2

Robinet, L., Heu-Thao, S., Arberet, L., Michelin, A., Belhadj, O., Radepont, M., & Lecouteux, S. (2024). Practices of the Mont Saint-Michel scriptorium in the use of parchment for manuscripts in the 11th century. *Journal of Cultural Heritage*, *67*, 452–460. https://doi.org/10.1016/j.culher.2024.03.013

Sood, V., & Chauhan, R. P. (2024). Quantum computing: Impact on energy efficiency and sustainability. *Expert Systems with Applications*, *255*, 124401. https://doi.org/10.1016/j.eswa.2024.124401

Tang, W., Martonosi, M., Jones, T., & Mullins, R. (2024). Distributed quantum computing via integrating quantum and classical computing. *Computer*, *57*(4), 131–136. https://doi.org/10.1109/MC.2024.3360569

Vyshedskiy, A. (2022). Language evolution is not limited to speech acquisition: a large study of language development in children with language deficits highlights the importance of the voluntary imagination component of language. *Research Ideas and Outcome Journal*, *8*(7109), 296–33. https://doi.org/10.3897/rio.8.e86401

Wan, X. (n.d.). From pictography to phoneticity: A Peircean hierarchy of written signs.

Wojahn, R. (2004). Cuneiform--the world's first writing. (Mesopotamian writing). In *AppleSeeds* (Vol. 7, Number 2, pp. 14). Cricket Media.

Yang, S., Bögels, B. W. A., Wang, F., Xu, C., Dou, H., Mann, S., Fan, C., & de Greef, T. F. A. (2024). DNA as a universal chemical substrate for computing and data storage. *Nature Reviews. Chemistry*, *8*(3), 179–194. https://doi.org/10.1038/s41570-024-00576-4

Zhang, Y., & Nguyen, N.-T. (2017). Magnetic digital microfluidics - A review. *Lab on a Chip*, *17*(6), 994–18. https://doi.org/10.1039/c7lc00025a